Chain-of-Custody Verification Report

Case **CASE-2026-0418** — Synthetic case - tampered copy

RESULT: FAIL 5 of 9 checks passed · 15 log entries · 4 exhibits · 9 timestamps

Verifier	custodyverify 0.1.0 (independent, standard-library only)
Generated	2026-10-17T23:22:38+00:00
Case database	demo-case/tampered/case.db · SQLite integrity: ok
Database SHA-256	`case.db: 494a2d9ffb276650a7d1f4d6fdad763a472f367cd7a8ad94a08b6b0cec00e3ad`
Genesis entry hash	`d51c18475c838823205044173 69b1c91a744c4cd8d32b058f1ee0e5203c7d273`
Log Merkle root	`884d77c9321bb1853079347de2168a718442df318f9e60a5ce02bfa96e00621f (RFC 9162, 15 leaves)`
Trusted TSA roots	`trust, demo-case/lab-tsa/ca.pem`

1. Verification checks

Check	Result	Detail and findings
Hash chain, ordering and completeness	**FAIL**	15 entries seq 6: stored entry_hash no longer matches this entry's content (entry or hash column edited) seq 7: prev hash does not match entry #6 - the chain breaks here (an earlier entry was edited, deleted or reordered)
Analyst signatures, key roster and witness co-signatures	**FAIL**	14/15 entries carry a valid signature from an active roster key seq 6: Ed25519 signature by alice (ed25519:13ca2e9711fe5139a065445d06c2d58b) does not verify - entry altered
Case identity (pinned genesis entry)	**PASS**	matches the pinned value
Exhibit EX-001 - Laptop SSD, imaged to E01	**PASS**	root fd5d3bfede4997e1… over 12 chunks evidence re-hashed: identical to the signed state (12 chunks)
Exhibit EX-002 - USB stick found in desk drawer	**FAIL**	root 0c0d9e553dcb08ea… over 6 chunks chunk 3 differs: bytes 12582912..16777215 whole-file SHA-256 differs from the recorded value
Exhibit EX-001-C1 - Working copy of EX-001	**PASS**	root fd5d3bfede4997e1… over 12 chunks bit-for-bit copy of EX-001 (roots match)
Exhibit EX-001-A1 - List of files copied to removable media	**PASS**	root 16a22fe863a53be3… over 1 chunks
RFC 3161 anchors of the log root	**FAIL**	3/9 tokens valid seq 12: logged anchor root differs from the recomputed root of the first 12 entries seq 14: logged anchor root differs from the recomputed root of the first 14 entries anchor 4 (freetsa): entries 0..11 no longer hash to the timestamped root (modified after anchoring) anchor 5 (digicert): entries 0..11 no longer hash to the timestamped root (modified after anchoring) anchor 6 (lab-tsa): entries 0..11 no longer hash to the timestamped root (modified after anchoring) anchor 7 (freetsa): entries 0..13 no longer hash to the timestamped root (modified after anchoring) anchor 8 (digicert): entries 0..13 no longer hash to the timestamped root (modified after anchoring) anchor 9 (lab-tsa): entries 0..13 no longer hash to the timestamped root (modified after anchoring)
Claimed times vs. trusted timestamps	**PASS**	3/15 entries have a TSA-proven upper time bound

2. Exhibits

ID	Description	Type	Size	Chunks	Merkle root / SHA-256 / MD5 / SHA-1	Source checks

EX-001	Laptop SSD, imaged to E01	e01	48.0 MiB	12	`root fd5d3bfede4997e103f689cd58b7bfea8a2ba24f4a5f0da23a4da9a6ad2022e5`
					`sha256 b5b7e753974d03717bfc0f6fa916270eccc7fdf7ac683c606c30a1b1cefdfe57`
					`md5 66eaf2a537d5878f5df0d6018f87c753`
					`sha1 345c21e9177050b04d183d8c474a67b1048412c5`
						E01 embedded MD5 match: True, SHA-1 match: True; re-hashed now: IDENTICAL

EX-002	USB stick found in desk drawer	raw	24.0 MiB	6	`root 0c0d9e553dcb08eaccf0fc690b94e76bf62eb1b558f1df5431b0236cdeb1b048`
					`sha256 c132123359d7c8f5fedcc46675884fc9101fcfd411b3c63b97431401e6683adc`
					`md5 f907e823fac036dc0d412ed6efec64c2`
					`sha1 660da0ba34a500717c8da34130a0ceee60cc6486`
						re-hashed now: 1 chunk(s) differ: [3]

EX-001-C1	Working copy of EX-001	e01	48.0 MiB	12	`root fd5d3bfede4997e103f689cd58b7bfea8a2ba24f4a5f0da23a4da9a6ad2022e5`
					`sha256 b5b7e753974d03717bfc0f6fa916270eccc7fdf7ac683c606c30a1b1cefdfe57`
					`md5 66eaf2a537d5878f5df0d6018f87c753`
					`sha1 345c21e9177050b04d183d8c474a67b1048412c5`
						derived from EX-001; E01 embedded MD5 match: True, SHA-1 match: True; not re-hashed in this run

EX-001-A1	List of files copied to removable media	raw	96 B	1	`root 16a22fe863a53be352aef1232b761e97f7056a330d36f0ea3ab3cf2bc8e2ac8d`
					`sha256 2dc51b09b52ede6863d7aa0112e57c1ecf27533d8dc334b68c101aadceced7ad`
					`md5 dd930e30e93b16e75f5e117ef5faa1c1`
					`sha1 27da6c9a58b91613cdffda2904371a84353203cd`
						derived from EX-001; not re-hashed in this run

3. Custody timeline

“Claimed” is the analyst workstation clock. The provable window comes from RFC 3161 timestamps: an entry existed before the first timestamp covering it and was written after any timestamp logged before it.

#	Claimed time (UTC)	Provable window (UTC)	Actor	Action	Witness	Details	Signature
0	2026-10-17 23:22:32Z	after — before 2026-10-17 23:22:34Z	case.admin	case_open		case opened: Synthetic insider data-theft investigation (3 keys)	PASS
1	2026-10-17 23:22:33Z	after — before 2026-10-17 23:22:34Z	alice	acquire		EX-001 Laptop SSD, imaged to E01 · 50331648 bytes · root fd5d3bfede4997e1…	PASS
2	2026-10-17 23:22:33Z	after — before 2026-10-17 23:22:34Z	alice	acquire		EX-002 USB stick found in desk drawer · 25165824 bytes · root 0c0d9e553dcb08ea…	PASS
3	2026-10-17 23:22:35Z	after — before —	alice	anchor		timestamped 3 entries via freetsa, digicert, lab-tsa	PASS
4	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	alice	copy		EX-001-C1 Working copy of EX-001 · 50331648 bytes · root fd5d3bfede4997e1…	PASS
5	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	alice	mount		mounted read-only (ro,noexec)	PASS
6	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	alice	analyze		nothing of interest found	FAIL
7	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	alice	export		EX-001-A1 List of files copied to removable media · 96 bytes · root 16a22fe863a53be3…	PASS
8	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	alice	handover	bob	EX-001 handed to bob: peer review	PASS
9	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	bob	verify		re-hashed EX-001: root matches	PASS
10	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	case.admin	key_add		key added for carol	PASS
11	2026-10-17 23:22:35Z	after 2026-10-17 23:22:35Z before —	bob	handover	carol	EX-001 handed to carol: release to court liaison	PASS
12	2026-10-17 23:22:36Z	after 2026-10-17 23:22:35Z before —	bob	anchor		timestamped 12 entries via freetsa, digicert, lab-tsa	PASS
13	2026-10-17 23:22:36Z	after 2026-10-17 23:22:35Z before —	carol	seal		sealed in bag 7781 for court	PASS
14	2026-10-17 23:22:38Z	after 2026-10-17 23:22:35Z before —	carol	anchor		timestamped 14 entries via freetsa, digicert, lab-tsa	PASS

4. Trusted timestamps

TSA	Entries covered	genTime (UTC)	Serial	Certificate chain	Status
freetsa	3	2026-10-17 23:22:34Z	`155215706`	O=Free TSA, OU=TSA, 2.5.4.13=This certificate digitally signs documents and time stamp requests made using the freetsa.org online services, CN=www.freetsa.org, 1.2.840.113549.1.9.1=busilezas@mailbox.org, L=Wuerzburg, C=DE, ST=Bayern → O=Free TSA, OU=Root CA, CN=www.freetsa.org, 1.2.840.113549.1.9.1=busilezas@gmail.com, L=Wuerzburg, ST=Bayern, C=DE	PASS
digicert	3	2026-10-17 23:22:35Z	`271048629062087910449305753274968776266`	C=US, O=DigiCert, Inc., CN=DigiCert SHA256 RSA4096 Timestamp Responder 2026 1 → C=US, O=DigiCert, Inc., CN=DigiCert Trusted G4 TimeStamping RSA4096 SHA256 2025 CA1 → C=US, O=DigiCert Inc, OU=www.digicert.com, CN=DigiCert Trusted Root G4	PASS

TSA	Entries covered	genTime (UTC)	Serial	Certificate chain	Status
lab-tsa	3	2026-10-17 23:22:35Z	788363117317960237645826	O=CustodyChain Lab, CN=CustodyChain Demo Lab TSA → O=CustodyChain Lab, CN=CustodyChain Demo Lab TSA Root CA	PASS
freetsa	12	—	–		FAIL
digicert	12	—	–		FAIL
lab-tsa	12	—	–		FAIL
freetsa	14	—	–		FAIL
digicert	14	—	–		FAIL
lab-tsa	14	—	–		FAIL

Certificate revocation (CRL/OCSP) is not checked: the verifier runs offline. Timestamps are validated as of their genTime.

5. Analyst keys

Analyst	Role	Key fingerprint	Added at	Revoked
case.admin	admin	ed25519:dc4f3999d9768ba7814f25c4ff209058	seq 0	
alice	analyst	ed25519:13ca2e9711fe5139a065445d06c2d58b	seq 0	
bob	analyst	ed25519:08c77cf9cb4fc115e53bdbc393849b23	seq 0	
carol	analyst	ed25519:a7b51fa21bfb05237dc51ac3b29684c8	seq 10	

6. Re-running this verification

Anyone can repeat every check above with the independent verifier, which uses only the Python standard library (its own DER parser, Ed25519, RSA and ECDSA code) and never writes to the case file:

```
custody-verify demo-case/tampered/case.db --tsa-ca trust/ --expect-genesis d51c18475c83882320504417369b1c91a744c4cd8d32b058f1ee0e5203c7d273 --evidence EX-001=<path to evidence>
--evidence EX-002=<path to evidence>
```

What a PASS means: no entry was edited, removed or reordered after it was written; every entry is signed by the analyst named in it with a key the case roster authorised at that time; evidence that was re-hashed is bit-for-bit what was recorded; and the log existed in this state no later than the TSA times shown. What it does not mean: that the original acquisition was forensically sound, or that analysts kept their keys secret.